Makanak Real Estate Platform

User Stories & Use Case Specifications

Project: Makanak Real Estate Platform
Project Type: Full-Stack .NET Real Estate Platform
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1. Actors

1.1 Visitor

An unauthenticated user who can browse public property listings and register for an account.

1.2 Buyer/Tenant

A registered user who can browse, search, filter, and sort property listings. The Buyer/Tenant can submit price offers, respond to counter-offers, and track offer statuses and negotiation history.

1.3 Owner

A registered user who can create and manage property listings, upload property images, set asking prices, and accept, reject, or submit counter-offers on their properties.

1.4 Admin

An authorized user responsible for managing user accounts, reviewing and managing property listings, and monitoring platform activity.

2. User Stories

2.1 Epic 1: Authentication and Authorization

US-01: User Registration

As a Visitor, I want to register an account, so that I can access the platform's features.

US-02: User Login

As a registered user, I want to log in securely, so that I can access my account and authorized features.

US-03: JWT Authentication

As a registered user, I want to receive a JWT after successful login, so that I can securely access protected API endpoints.

US-04: Role-Based Authorization

As a user, I want the system to enforce role-based permissions, so that I can perform only the actions authorized for my role.

US-05: Resource Access Protection

As a user, I want the system to prevent unauthorized access to or modification of resources belonging to other users, so that my data remains protected.

2.2 Epic 2: Property Management

US-06: Create Property Listing

As an Owner, I want to create a property listing with an asking price and listing type, so that I can advertise my property for sale or rent.

US-07: Upload Property Images

As an Owner, I want to upload property images and select a primary image, so that potential buyers and tenants can view my property clearly.

US-08: Update Property Listing

As an Owner, I want to update my property details, so that the listing remains accurate.

US-09: Delete Property Listing

As an Owner, I want to remove my property listing, so that I can stop advertising a property I no longer wish to list.

US-10: Manage Property Status

As an Owner, I want to manage my property's status, so that users know whether it is available, sold, rented, or inactive.

2.3 Epic 3: Property Search and Filtering

US-11: Browse Properties

As a Buyer/Tenant, I want to browse available property listings, so that I can explore suitable options.

US-12: Search Properties

As a Buyer/Tenant, I want to search for properties using keywords and city names, so that I can find relevant listings.

US-13: Filter Properties

As a Buyer/Tenant, I want to filter properties by price range, property type, listing type, and city, so that I can narrow down my search results.

US-14: Sort and Paginate Results

As a Buyer/Tenant, I want to sort listings by price or creation date and navigate through paginated results, so that I can browse properties efficiently.

2.4 Epic 4: Offers and Negotiation

US-15: Submit a Price Offer

As a Buyer/Tenant, I want to submit a price offer on an available property I do not own, so that I can negotiate with the Owner.

US-16: Accept or Reject an Offer

As an Owner, I want to accept or reject an offer submitted for my property, so that I can decide whether the proposed price is acceptable.

US-17: Submit a Counter-Offer

As an Owner, I want to submit a counter-offer with a different price, so that I can propose a price that better meets my expectations.

US-18: Respond to a Counter-Offer

As a Buyer/Tenant, I want to accept, reject, or submit another counter-offer, so that I can continue negotiating the property price.

US-19: View Negotiation History

As a user involved in an offer, I want to view its negotiation history, so that I can track previous actions, proposed prices, and timestamps.

US-20: Validate Offers and Prevent Conflicts

As a user, I want the system to validate offers and prevent conflicting acceptances, so that the negotiation process remains consistent and reliable.

2.5 Epic 5: User Dashboards

US-21: Owner Dashboard

As an Owner, I want to manage my property listings and review received offers through a dashboard, so that I can manage my properties and negotiations in one place.

US-22: Buyer/Tenant Dashboard

As a Buyer/Tenant, I want to view my submitted offers and their current statuses through a dashboard, so that I can track my negotiations.

2.6 Epic 6: Administration

US-23: Manage User Accounts

As an Admin, I want to view user accounts and manage accounts that violate platform rules, so that I can maintain a safe and reliable platform.

US-24: Manage Property Listings

As an Admin, I want to review property listings and deactivate or remove listings that violate platform rules, so that inappropriate or misleading content can be addressed.

US-25: Monitor Platform Activity

As an Admin, I want to view relevant platform activity and basic system information, so that I can monitor the platform and identify potential issues.

3. Use Case Specifications

UC-01: Register Account

Primary Actor: Visitor
Preconditions:
The visitor is not logged in.
The visitor does not already have an account using the same unique registration identifier.
Main Flow:

- 1. The visitor opens the registration page.
- 2. The visitor enters the required registration details.
- 3. The system validates the submitted information.
- 4. The system creates the account and securely stores the user's credentials.
- 5. The system displays a success message.

Alternative Flows:
If required fields are missing or invalid, the system displays validation errors.
If the registration identifier is already in use, the system rejects the registration and displays an appropriate message.

Postconditions:
The user account is created successfully.

UC-02: Log In and Access Protected Resources

Primary Actor: Registered User
Preconditions:
The user has a registered account.
Main Flow:

- 1. The user opens the login page.
- 2. The user enters their credentials.
- 3. The system validates the credentials.
- 4. The system issues a JWT after successful authentication.
- 5. The user includes the token in requests to protected API endpoints.
- 6. The system validates the token.
- 7. The system checks the user's role and permissions before allowing the requested operation.

Alternative Flows:
If the credentials are invalid, the system rejects the login attempt.
If the token is missing, invalid, or expired, the system denies access to the protected resource.
If the user does not have the required permissions, the system denies the operation.
Postconditions:
The authenticated user can access only resources and operations authorized for their role.

UC-03: Create Property Listing

Primary Actor: Owner
Preconditions:
The Owner is authenticated.
The Owner has permission to create property listings.
Main Flow:

- 1. The Owner opens the dashboard.
- 2. The Owner selects Add Property.
- 3. The Owner enters the property title, description, price, address, and city.
- 4. The Owner selects the property type, such as Apartment, Villa, or Office.
- 5. The Owner selects the listing type: Sale or Rent.
- 6. The Owner uploads property images and selects a primary image.
- 7. The system validates the submitted information.
- 8. The system saves the listing and associates it with the Owner.
- 9. The system assigns the initial property status according to the business rules.
- 10. The system displays a success message.

Alternative Flows:
If required data is missing or invalid, the system displays validation errors.

If image uploading fails, the system displays an appropriate error message.
Postconditions:
The property listing is saved with its details, asking price, listing type, images, and Owner association.

UC-04: Search and Filter Properties

Primary Actor: Buyer/Tenant
Preconditions:
The property listings page is accessible.
Main Flow:

- 1. The user opens the property listings page.
- 2. The user enters keywords or selects a city.
- 3. The user applies filters for price range, property type, and listing type.
- 4. The user selects a sorting option if needed.
- 5. The system retrieves matching property listings.
- 6. The system displays paginated results with property details and primary images.
- 7. The user opens a listing to view its full details and asking price.

Alternative Flows:
If no properties match the criteria, the system displays an appropriate message.
If the search criteria change, the system retrieves updated results.
Postconditions:
The user can view properties matching the selected criteria.

UC-05: Submit a Price Offer

Primary Actor: Buyer/Tenant
Preconditions:

- The user is authenticated.

The property is available.

- The user does not own the property.

Main Flow:

- 1. The user opens an available property listing.
- 2. The user reviews the property details and asking price.
- 3. The user selects Submit Offer.
- 4. The user enters a proposed price.
- 5. The system validates the offer price and the property's current status.
- 6. The system creates the offer and associates it with the user and property.
- 7. The system records the initial negotiation action, including the user, proposed price, and timestamp.
- 8. The system makes the offer available to the Owner for review.

Alternative Flows:

- If the proposed price is invalid, the system rejects the offer and displays a validation message.

If the property is no longer available, the system prevents the offer from being submitted.
If the user owns the property, the system rejects the operation.
Postconditions:
The offer is saved and available for the Owner to review.

UC-06: Respond to an Offer and Negotiate

Primary Actors: Owner and Buyer/Tenant
Preconditions:
An offer exists for a property.

- The current user is authorized to respond to the offer.

The offer is in a state that allows the requested action.
Main Flow:

- 1. The Owner opens the received offers section.
- 2. The Owner selects an offer to review.
- 3. The Owner chooses Accept, Reject, or Counter-Offer.
- 4. If the Owner chooses Counter-Offer, the Owner enters a new proposed price.
- 5. The system validates the action, proposed price, and current offer state.
- 6. The system updates the offer status as appropriate.
- 7. The system records the action, acting user, proposed price, and timestamp in the negotiation history.
- 8. If a counter-offer is submitted, the Buyer/Tenant can review it.
- 9. The Buyer/Tenant can accept, reject, or submit another counter-offer, provided the offer remains eligible for a response.
- 10. The system updates the offer status and negotiation history after each valid action.

Alternative Flows:

- If the offer no longer allows the requested action, the system rejects the operation.

If the property becomes unavailable, the system prevents acceptance according to the business rules.
If another offer has already been accepted for the same property, the system prevents a conflicting acceptance.
If the proposed price is invalid, the system displays a validation error.
Postconditions:

- The offer status reflects the latest valid action.

The negotiation history contains the recorded negotiation actions.

UC-07: View User Dashboard

Primary Actors: Owner and Buyer/Tenant
Preconditions:
The user is authenticated.
The user is authorized to access the dashboard.
Main Flow:

- 1. The user opens the dashboard.
- 2. The system identifies the user's role.
- 3. If the user is an Owner, the system displays their property listings and received offers.
- 4. If the user is a Buyer/Tenant, the system displays their submitted offers and current statuses.
- 5. The user selects an offer to view its details and negotiation history.

Alternative Flows:
If no relevant data exists, the system displays an appropriate empty-state message.
If the user attempts to access another user's restricted information, the system denies access.
Postconditions:

- The user views only the dashboard information authorized for their role.

UC-08: Manage User Accounts

Primary Actor: Admin
Preconditions:
The Admin is authenticated.
The Admin has the required administrative permissions.
Main Flow:

- 1. The Admin opens the administration dashboard.
- 2. The Admin selects User Management.
- 3. The system displays user accounts and permitted account information.
- 4. The Admin selects a user account.
- 5. The Admin reviews the account details.
- 6. The Admin performs an authorized action, such as deactivating an account that violates platform rules.
- 7. The system validates the action and updates the account status.
- 8. The system confirms the result.

Alternative Flows:
If the selected account does not exist, the system displays an appropriate error.
If the Admin lacks the required permission, the system denies the operation.
If the requested action is invalid, the system rejects it.
Postconditions:
The account is updated only when the requested action is valid and authorized.

UC-09: Review and Manage Property Listings

Primary Actor: Admin
Preconditions:

- The Admin is authenticated and authorized.
- Property listings exist in the system.

Main Flow:

- 1. The Admin opens the administration dashboard.
- 2. The Admin selects Property Management.

- 3. The system displays property listings and relevant details.
- 4. The Admin selects a listing for review.
- 5. The Admin reviews the listing information.
- 6. If the listing violates platform rules, the Admin chooses an authorized action, such as deactivating or removing it.
- 7. The system validates the action and applies the change.
- 8. The system confirms the result.

Alternative Flows:
If the listing does not exist, the system displays an appropriate error.
If the Admin lacks permission, the system denies the operation.
If the action would violate data integrity rules, the system rejects it.
Postconditions:
The listing is unchanged or updated according to the authorized action.
Related offer and negotiation data is handled according to the agreed data-retention rules.

UC-10: Monitor Platform Activity

Primary Actor: Admin
Preconditions:
The Admin is authenticated and authorized to view platform information.
Main Flow:

- 1. The Admin opens the administration dashboard.
- 2. The Admin selects Platform Monitoring.
- 3. The system displays available platform activity records and basic system information.
- 4. The Admin reviews the displayed information.
- 5. The Admin identifies potential issues that may require investigation.

Alternative Flows:

- If no activity records are available, the system displays an appropriate message.

If the Admin lacks permission to access the information, the system denies access.
Postconditions:
The Admin can review available platform information within their authorized permissions.

4. User Story and Use Case Mapping

5. Business Rules and Constraints

UC-01	Register Account	US-01

UC-02	Log In and Access Protected Resources	US-02, US-03, US-04, US-05
UC-03	Create Property Listing	US-06, US-07
UC-04	Search and Filter Properties	US-11, US-12, US-13, US-14
UC-05	Submit a Price Offer	US-15, US-20
UC-06	Respond to an Offer and Negotiate	US-16, US-17, US-18, US-19, US-20
UC-07	View User Dashboard	US-21, US-22
UC-08	Manage User Accounts	US-23
UC-09	Review and Manage Property Listings	US-24
UC-10	Monitor Platform Activity	US-25

- 1. Authentication is required for protected operations.
- 2. Authorization must be enforced on the server side, not only in the user interface.
- 3. User roles are Buyer/Tenant, Owner, and Admin.
- 4. Owners can manage only their own property listings and associated offers.
- 5. Buyers/Tenants cannot submit offers on properties they own.
- 6. Offers can only be submitted for available properties.
- 7. Offer prices must satisfy the system's validation rules.
- 8. The system must prevent conflicting acceptances for the same property, including concurrent requests.
- 9. Negotiation actions must be recorded with the acting user, action, proposed price, and timestamp.
- 10. Admin permissions include managing user accounts, managing property listings, and monitoring platform activity.
- 11. Administrative actions must be authorized and validated.
- 12. Related offers and negotiation history must be handled according to the agreed data-integrity and retention rules.

- 13. Real-time notifications, online payment gateways, interactive maps, email/SMS notifications, advanced analytics, and direct messaging outside the offer negotiation process are excluded from the current MVP scope.

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